

Software supports the functioning of a computer system internally and cannot be seen. It is stored on secondary memory and can be an **application software** as well as **system software**. The application software is used to perform a specific task according to requirements and the system software (operating system and networking system) is mandatory for running application software.

## 1.5 INPUT DEVICES

**LO 1.3**

Input devices are electromechanical devices that are used to provide data to a computer for storing and further processing, if necessary. Depending upon the type or method of input, the input device may belong to one of the following categories:

### 1.5.1 Keyboard

Keyboard is used to type data and text and execute commands. A standard keyboard, as shown in Fig. 1.8, consists of the following groups of keys:

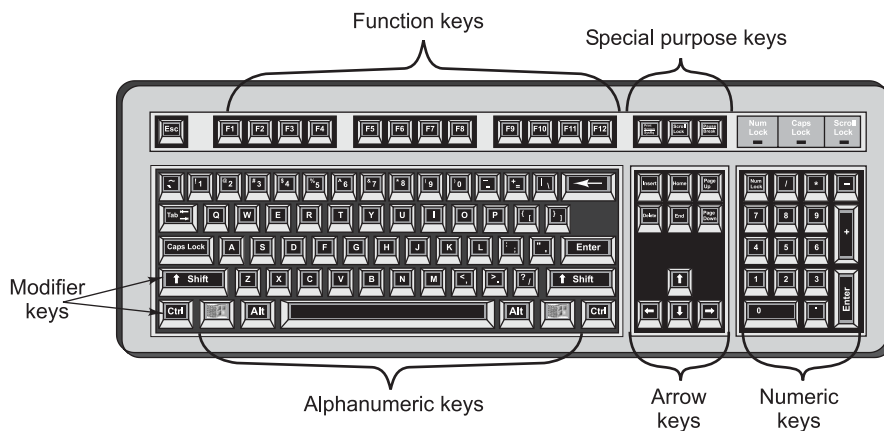
**Alphanumeric Keys** include the number keys and alphabet keys arranged in QWERTY layout.

**Function Keys** help perform specific tasks, such as searching a file or refreshing a web page.

**Central Keys** include arrow keys (for moving the cursor) and modifier keys such as SHIFT, ALT and CTRL (for modifying the input).

**Numeric Keypad** looks like a calculator's keypad with its 10 digits and mathematical operators.

**Special Purpose Keys** The special purpose keys help perform a certain kind of operation, like exiting a program (Escape) or deleting some characters (Delete) in a document, etc.

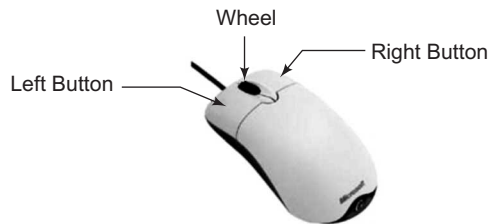


**Fig. 1.8** The presently used keyboard

### 1.5.2 Mouse

Mouse is a small hand-held pointing device that basically controls the two-dimensional movement of the cursor on the displayed screen. It is an important part of the Graphical User Interface (GUI) based Operating Systems (OS) as it helps in selecting a portion of the screen and copying and pasting the text.

The mouse, on moving, also moves the pointer appearing on the display device (Fig. 1.9).



**Fig. 1.9** A mechanical mouse

### 1.5.3 Scanning Device

Scanning devices are the input devices that can electronically capture text and images, and convert them into computer readable form (Fig. 1.10).

There are the following types of scanners that can be used to produce digitized images:

- ❖ **Flatbed scanner** — It contains a scanner head that moves across a page from top to bottom to read the page and converts the image or text available on the page in digital form. The flatbed scanner is used to scan graphics, oversized documents, and pages from books.
- ❖ **Drum scanner** — In this type of scanner, a fixed scanner head is used and the image to be scanned is moved across the head. The drum scanners are used for scanning prepress materials.
- ❖ **Slide scanner** — It is a scanner that can scan photographic slides directly to produce files understandable by the computer.
- ❖ **Handheld scanner** — It is a scanner that is moved by the end user across the page to be scanned. This type of scanner is inexpensive and small in size.



**Fig. 1.10** A Scanner

## 1.6 PROCESSOR

### LO 1.3

The CPU consists of Control Unit (CU) and ALU. CU stores the instruction set, which specifies the operations to be performed by the computer. CU transfers the data and the instructions to the ALU for an arithmetic operation. ALU performs arithmetical or logical operations on the data received. The CPU registers store the data to be processed by the CPU and the processed data also. Apart from CU and ALU, CPU seeks help from the following hardware devices to process the data:

#### Motherboard

It refers to a device used for connecting the CPU with the input and output devices. The components on the motherboard are connected to all parts of a computer and are kept insulated from each other. Some of the components of a motherboard are:

- ❖ **Bus:** Electrical pathways that transfer data and instructions among different parts of the computer. For example, the data bus is an electrical pathway that transfers data among the microprocessor, memory and input/output devices connected to the computer.
- ❖ **System clock:** It is a clock used for synchronizing the activities performed by the computer. The electrical signals that are passed inside a computer are timed, based on the tick of the clock.
- ❖ **Microprocessor:** CPU component that performs the processing and controls the activities performed by the different parts of the computer.
- ❖ **ROM:** Chip that contains the permanent memory of the computer that stores information, which cannot be modified by the end user.

## RAM

It refers to primary memory of a computer that stores information and programs, until the computer is used. RAM is available as a chip that can be connected to the RAM slots in the motherboard.

## Video Card/Sound Card

The video card is an interface between the monitor and the CPU. Video cards also include their own RAM and microprocessors that are used for speeding up the processing and display of a graphic. A sound card is a circuit board placed on the motherboard and is used to enhance the sound capabilities of a computer.

## 1.7 OUTPUT DEVICES

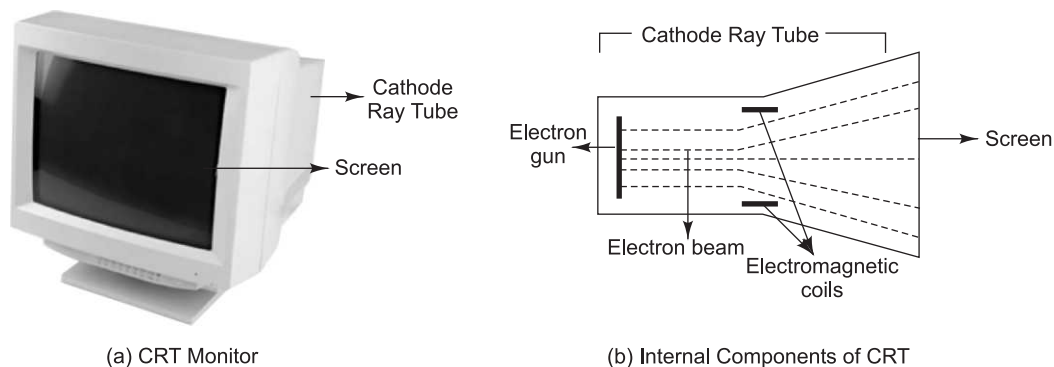
**LO 1.3**

The main task of an output device is to convert the machine-readable information into human-readable form which may be in the form of text, graphics, audio or video.

### 1.7.1 Display Monitors

A monitor produces visual displays generated by the computer. The monitor is connected to the video card placed on the expansion slot of the motherboard.

The monitors can be classified as cathode ray tube (CRT) monitors or liquid crystal display (LCD) monitors. The CRT monitors are large, occupy more space in the computer, whereas LCD monitors are thin, light weighted, and occupy lesser space. Both the monitors are available as monochrome, gray scale and color models.



**Fig. 1.11** A CRT monitor and the internal components of a CRT

A monitor can be characterized by its monitor size and resolution. The monitor size is the length of the screen that is measured diagonally. The resolution of the screen is expressed as the number of picture elements or pixels of the screen. The resolution of the monitor is also called the dot pitch. The monitor with a higher resolution produces a clearer image.

### 1.7.2 Printer

The printer is an output device that transfers the text displayed on the screen, onto paper sheets that can be used by the end user. Printers can be classified based on the technology they use to print the text and images:

- ❖ **Dot matrix printers** — Dot matrix printers are impact printers that use perforated sheet to print the text. Dot matrix printers are used to produce multiple copies of a print out.
- ❖ **Inkjet printers** — Inkjet printers are slower than dot matrix printers and are used to generate high quality photographic prints.
- ❖ **Laser printers** — The laser printer may or may not be connected to a computer, to generate an output. These printers consist of a microprocessor, ROM and RAM, which can be used to store the textual information.

### 1.7.3 Voice Output Systems

These systems record the simple messages in human speech form and then combine all these simple messages to form a single message. The voice response system is of two types—one uses a reproduction of human voice and other sounds, and the other uses speech synthesis.

The basic application of a voice output system is in Interactive Voice Response (IVR) systems, which are used by the customer care or customer support departments of an organization, such as telecommunication companies, etc.

### 1.7.4 Projectors

A projector is a device that is connected to a computer or a video device for projecting an image from the computer or video device onto the big white screen. The images projected by a projector are larger in size as compared to the original images. A projector consists of an optic system, a light source and displays, which contain the original images. Projectors were initially used for showing films but now they are used on a large scale for displaying presentations in business organizations and for viewing movies at home.



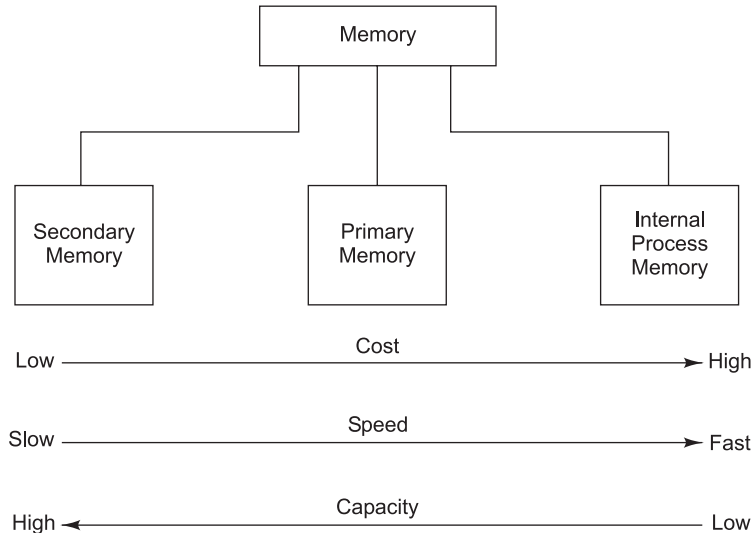
Fig. 1.12 A portable projector

## 1.8 MEMORY MANAGEMENT

LO 1.3

The memory unit of a computer is used to store data, instructions for processing data, intermediate results of processing and the final processed information. The memory units of a computer are classified as primary and secondary memory. Computers also use a third type of storage location known as the *internal process*

*memory*. This memory is placed either inside the CPU or near the CPU (connected through special fast bus).



**Fig. 1.13** Memory unit categories of computer

### 1.8.1 Primary Memory

The primary memory is available in the computer as a built-in unit of the computer. The primary memory is represented as a set of locations with each location occupying 8 bits. Each bit in the memory is identified by a unique address. The data is stored in the machine-understandable binary form in these memory locations. The commonly used primary memories are as follows:

- ❖ **ROM** — ROM represents Read Only Memory that stores data and instructions, even when the computer is turned off. It is the permanent memory of the computer where the contents cannot be modified by an end user. ROM is a chip that is inserted into the motherboard. It is generally used to store the Basic Input/Output system (BIOS), which performs the Power On Self Test (POST).
- ❖ **RAM** — RAM is the read/write memory unit in which the information is retained only as long as there is a regular power supply. When the power supply is interrupted or switched off, the information stored in the RAM is lost. RAM is volatile memory that temporarily stores data and applications as long as they are in use. When the use of data or the application is over, the content in RAM is erased.
- ❖ **Cache memory** — Cache memory is used to store the data and the related application that was last processed by the CPU. When the processor performs processing, it first searches the cache memory and then the RAM, for an instruction. The cache memory can be either soldered into the motherboard or is available as a part of RAM.

### 1.8.2 Secondary Memory

Secondary memory represents the external storage devices that are connected to the computer. They provide a non-volatile memory source used to store information that is not in use currently. A storage device is either located in the CPU casing of the computer or is connected externally to the computer. The secondary storage devices can be classified as:

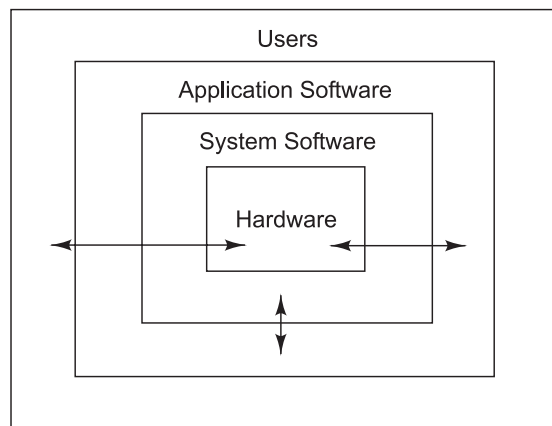
- ❖ **Magnetic storage device** — The magnetic storage devices store information that can be read, erased and rewritten a number of times. These include floppy disk, hard disk and magnetic tapes.
- ❖ **Optical storage device** — The optical storage devices are secondary storage devices that use laser beams to read the stored data. These include CD-ROM, rewritable compact disk (CD-RW), digital video disks with read only memory (DVD-ROM), etc.
- ❖ **Magneto-optical storage device** — The magneto-optical devices are generally used to store information, such as large programs, files and back-up data. The end user can modify the information stored in magneto-optical storage devices multiple times. These devices provide higher storage capacity as they use laser beams and magnets for reading and writing data to the device.

## 1.9 TYPES OF COMPUTER SOFTWARE

**LO 1.4**

A computer program is basically a set of logical instructions, written in a computer programming language that tells the computer how to accomplish a task. The software is therefore an essential interface between the hardware and the user (Fig. 1.14).

A computer software performs two distinctive tasks. The first task is to control and coordinate the hardware components and manage their performances and the second one is to enable the users to accomplish their required tasks. The software that is used to achieve the first task is known as the *system software* and the software that is used to achieve the second task is known as the *application software*.



**Fig. 1.14** Layers of software and their interactions

### 1.9.1 System Software

System software consists of many different programs that manage and support different tasks. Depending upon the task performed, the system software can be classified into two major groups (Fig. 1.15):

- ❖ *System management programs* used for managing both the hardware and software systems. They include:
  - Operating system
  - Utility programs
  - Device drivers

- ❖ *System development programs* are used for developing and executing application software. These are:
  - Language translators
  - Linkers
  - Debuggers
  - Editors

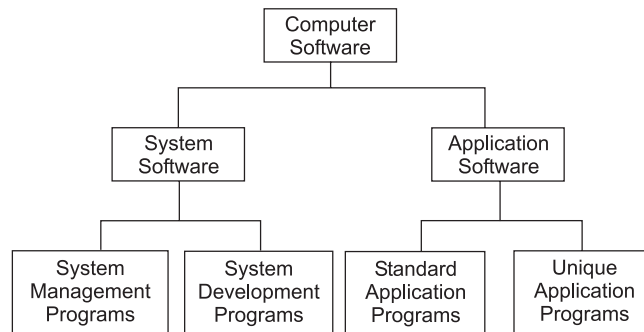


Fig. 1.15 Major categories of computer software

### 1.9.2 Application Software

Application software includes a variety of programs that are designed to meet the information processing needs of end users. They can be broadly classified into two groups:

- ❖ *Standard application programs* that are designed for performing common application jobs. Examples include:
  - Word processor
  - Spreadsheet
  - Database Manager
  - Desktop Publisher
  - Web Browser
- ❖ *Unique application programs* that are developed by the users themselves to support their specific needs. Examples include:
  - Managing the inventory of a store
  - Preparing pay-bills of employees in an organization
  - Reserving seats in trains or airlines

## 1.10 OVERVIEW OF OPERATING SYSTEM

LO 1.5

An operating system (OS) is a software that makes the computer hardware to work. While the hardware provides 'raw computer power', the OS is responsible for making the computer power useful for the users. OS is the main component of system software and therefore must be loaded and activated before we can accomplish any other task. The main functions include:

- ❖ Operates CPU of the computer.
- ❖ Controls input/output devices that provide the interface between the user and the computer.
- ❖ Handles the working of application programs with the hardware and other software systems.